

Jiaxuan Zou

Undergraduate Student in Mathematics and Statistics

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Undergraduate in Mathematics and Statistics at Xi'an Jiaotong University and research intern at the Gaoling School of Artificial Intelligence, Renmin University of China. My work focuses on mechanistic interpretability, training dynamics, optimizer design, and scaling laws.

Education

Xi'an Jiaotong University

2024 – Present

Undergraduate Student, Mathematics and Statistics

Xi'an, China

- School of Mathematics and Statistics.
- Research interests include deep learning theory, optimization, mechanistic interpretability, and scaling laws.

Experience

Gaoling School of Artificial Intelligence, Renmin University of China

Research Intern — Advised by Prof. Yong Liu

Beijing, China

- Work on mechanistic interpretability, deep learning theory, optimizer design, and scaling laws.
- Co-authored preprints on linear attention, neural scaling laws, latent chain-of-thought, and matrix optimizer design.

ScalingOpt

2026 – Present

Co-maintainer — Optimizer design for LLM training

Remote

- Work on the relation among optimizer design, model architecture, and training configuration under scaling-law regimes.
- Help maintain the optimizer library, including the Nora optimizer entry.

Tsinghua-affiliated AI Startup

AI Technical Consultant

Beijing, China

- Consult on “AI + K-12 Education” products developed with the Beijing Dongcheng District Education Commission.

Publications

Author names in **bold** indicate myself.

1. **Jiaxuan Zou**, Ruifeng Ren, Yong Liu. *Kaczmarz Linear Attention*. Under Review, 2026. [arXiv:2605.08587](#).
2. Jinghui Yuan, **Jiaxuan Zou**, Shuo Wang, Yong Liu, Feiping Nie. *Nora: Normalized Orthogonal Row Alignment for Scalable Matrix Optimizer*. Under Review, 2026. [arXiv:2605.03769](#).
3. **Jiaxuan Zou**, Zixuan Gong, Ye Su, Huayi Tang, Yong Liu. *Effective Frontiers: A Unification of Neural Scaling Laws*. Under Review, 2026. [arXiv:2602.02593](#).
4. **Jiaxuan Zou**, Yaozhong Xiong, Yong Liu. *Capabilities and Fundamental Limits of Latent Chain-of-Thought*. Under Review, 2026. [arXiv:2602.01148](#).
5. Jialong Sun, Zeming Wei, **Jiaxuan Zou**, Jiacheng Gong, Guanheng Wang, Chengyang Dong, Jialong Li, Bo Liu. *Statistical MIA: Rethinking Membership Inference Attack for Reliable Unlearning Auditing*. Under Review, 2026. [arXiv:2602.01150](#).
6. Jialong Sun, Xinpeng Ling, **Jiaxuan Zou**, Jiawen Kang, Kejia Zhang. *FreIE: Low-Frequency Spectral Bias in Neural Networks for Time-Series Tasks*. IEEE ICDM, 2025. [arXiv:2510.25800](#).

7. Tian Zhang, Jiaju He, Naihao Liu, **Jiaxuan Zou**, Yongxiang Jiang. *Mamba-Driven and Feature-Fused U-Net for Automatic Seismic Horizon Interpretation*. IEEE Transactions on Geoscience and Remote Sensing, 2026. doi:10.1109/TGRS.2026.3660880.
8. **Jiaxuan Zou**, Naihao Liu, Tian Zhang, Jiaju He, Tao Li, Jinghuai Gao. *Seismic Horizon Picking Using Channel-Independent Multi-Scale UNet*. Under Review, 2024.

Projects

ScalingOpt

tianshijing.github.io/ScalingOpt

- Co-maintainer of a discussion platform and benchmark project for optimizer design in LLM training.
- Focuses on optimizer design, model architecture, and training configuration under the scaling-law paradigm.
- The Nora optimizer has been included in the ScalingOpt optimizer library.

Research Interests

- Mechanistic interpretability of LLMs.
- Training dynamics of finite-width neural networks.
- Optimizer design for LLM pre-training.
- Scaling laws and their failure modes.
- Deep learning theory and optimization.

Skills & Languages

Research areas: Mechanistic interpretability, Deep learning theory, Optimization, Scaling laws, Training dynamics.

Mathematical tools: Optimization, Statistics, Dynamical systems, Mathematical modeling.

Applied AI: LLM pre-training, Matrix optimizers, Linear attention, AI education products.

Languages: Chinese (Native), English (Working proficiency).